

CORE RDK BROADBAND PLATFORM

Hardware Compatibility Specification

Minimum Hardware Configuration per Device Profile

Document Status	R1 — Baseline for Review
Owner	RDKM
Scope	2 of 7 RDK-B device profiles: EthWAN WiFi Router, EXT EasyMesh (see Section 1 for all profiles defined)
Reference Platform	BananaPi BPI-R4 (MediaTek MT7988 / Filogic), Kirkstone 4.0, Kernel 6.6
Related Work	Core RDK Broadband
Last Updated	17 September 2026

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1. Purpose and Scope

This document defines the minimum hardware configuration required to run the Core RDK Broadband (RDK-B) platform for each supported device profile. It establishes a hardware compatibility baseline covering minimum CPU, memory, and flash storage, along with the peripheral hardware required to support the software and connectivity features of each profile.

The specification is intended to be used by silicon vendors, ODMs, and platform integrators to evaluate whether a candidate hardware design is viable for Core RDK Broadband, and by the RDK-B architecture team as the baseline against which reference devices are validated.

1.1 Device Profiles Defined for RDK-B

The RDK-B Component List (2026) defines seven device profiles spanning the full range of gateway, extender, and bridge deployments. All seven are listed below for completeness and traceability.

Device Profile	Coverage in this Spec	Definition (RDK-B Component List 2026)
Modem / ONU	Defined — not covered in this revision	Managed or unmanaged bridge from a WAN access technology to an Ethernet LAN port(s); may provide WAN configuration and diagnostics data. Does not include routing or Wi-Fi AP features.
EthWAN WiFi Router	In scope (this document)	IPv4/IPv6 router with Ethernet WAN and Wi-Fi AP features. EasyMesh features are optional.
GW	Defined — not covered in this revision	IPv4/IPv6 router with one or more WAN access technologies and Wi-Fi AP features.
GW OpenSync	Defined — not covered in this revision	IPv4/IPv6 router with one or more WAN access technologies and OpenSync Mesh Wi-Fi features.
GW EasyMesh	Defined — not covered in this revision	IPv4/IPv6 router with one or more WAN access technologies and EasyMesh Wi-Fi controller + agent features.
EXT OpenSync	Defined — not covered in this revision	Wi-Fi extender device based on OpenSync technology; supports Wi-Fi AP fronthaul and Wi-Fi/Ethernet backhaul.
EXT EasyMesh	In scope (this document)	Wi-Fi extender device based on EasyMesh Agent technology; supports Wi-Fi AP fronthaul and Wi-Fi/Ethernet backhaul.

1.2 Scope of This Specification

This revision of the Hardware Compatibility Specification covers two device profiles only: (1) EthWAN WiFi Router, and (2) EXT EasyMesh. The remaining five profiles (Modem/ONU, GW, GW OpenSync, GW EasyMesh, EXT OpenSync) are defined above for context but are not yet covered by minimum CPU/memory/flash or peripheral requirements in this document; they are candidates for a future revision.

- EthWAN WiFi Router: an IPv4/IPv6 router with a dedicated Ethernet WAN uplink and Wi-Fi AP features; EasyMesh is optional on this profile rather than mandatory.

- EXT EasyMesh: a Wi-Fi extender/satellite device operating as an EasyMesh R6 agent only, with Wi-Fi AP fronthaul and Wi-Fi or Ethernet backhaul to a controller.

1.3 Normative Keywords

The keywords MUST, MUST NOT, REQUIRED, SHALL, SHALL NOT, SHOULD, SHOULD NOT, RECOMMENDED, MAY, and OPTIONAL in this document are to be interpreted per RFC 2119: MUST/REQUIRED/SHALL denote an absolute requirement; MUST NOT/SHALL NOT denote an absolute prohibition; SHOULD/RECOMMENDED denote a requirement that may be deviated from with documented justification; MAY/OPTIONAL denote a genuinely optional item.

Requirement tables in this specification use "Required"/"Optional"/"Not applicable" as the tabular shorthand for MUST/MAY/not-in-scope respectively, consistent throughout Sections 3–4 and 9. Every requirement carries a stable identifier (e.g., CA-001, PER-003) so it can be cited from a test case, an errata entry, or a PICS declaration (Section 9) without ambiguity — identifiers are assigned once per revision and are not reused if a requirement is removed.

Identifier prefixes: CRF (CPU/RAM/Flash floor, Section 3), CA (Compute Architecture, 3.4), MEM (Memory, 3.5), FLC (Flash Logical Class, 4.3), STG (Storage, 4.3), PER (top-level Peripherals, Section 4), PCH (Platform-Control Hardware, 4.2), NET (Network Hardware, 4.4), WIFI (Wi-Fi Hardware, 4.5).

2. Referenced Documents

- RDK-B Component List 2026 — authoritative source for device profile definitions and per-profile Required/Optional/N-A feature mapping used throughout this document.

3. Minimum CPU, Memory, and Flash Storage

Before the numbers below: what do “CPU,” “RAM,” and “Flash” actually have to cover? A production broadband device's software stack has multiple layers, not just Core RDK-B — shown here for a representative production gateway build:

Software Stack Layer
Core RDK-B Components
SoC-Specific Drivers & HAL
OEM Components (FW upgrade, etc.)
ISP Services & Configuration
Vendor Apps (e.g., Cujo, Speed Test)
Cloud Management (e.g., OpenSync)
Home Automation Services
Additional Protocol Support

What RAM Must Cover

RAM must hold, concurrently: the Linux kernel and its loaded drivers; the open-source (OSS) user-space runtime the system depends on (Python, OpenSSL, database and media libraries, and similar packages — see the OSS breakdown in Section 3.6); Core RDK-B's own management processes (CCSP stack, OneWifi/EasyMesh, WebPA/TR-069/TR-369

agents, telemetry); and, on a fully-provisioned device, whatever OEM firmware-upgrade tooling, ISP-specific services, vendor apps, cloud-management agents, or home-automation services the operator adds on top.

What Flash Must Cover

Flash must store: the bootloader and kernel image (in A/B pairs — see Section 3.6's A/B banking principle and Section 4.3); kernel modules; the OSS packages the runtime depends on; Core RDK-B's own rootfs; and, again, whatever OEM/ISP/vendor/cloud layers are added. Section 3.6 gives the exact measured OSS/OEM/kernel/RDK-B breakdown for both in-scope profiles' actual base images.

What CPU Must Cover

CPU must have enough headroom to run all of the above concurrently — kernel scheduling and interrupt handling, driver overhead, OSS runtime execution, Core RDK-B's own processes, and any additional vendor/operator workload — without starving any single layer, particularly under load (active Wi-Fi clients, sustained WAN/backhaul throughput, EasyMesh formation and steering).

The CPU/RAM/Flash figures in the table below cover the Linux kernel, SoC drivers/HAL, the open-source (OSS) runtime, and Core RDK-B components together — measured and evidenced in Sections 5 and 3.7 — not a complete device budget. OEM components, ISP services, vendor apps, and cloud/home-automation layers are additional and must be sized on top of these numbers by whoever adds them. See Section 3.3 for the full scope boundary and Section 3.6 for the layer-by-layer composition model and measured OSS/OEM/RDK-B breakdown.

The table below defines the minimum viable hardware configuration for the two in-scope profiles: EthWAN WiFi Router and EXT EasyMesh. These values represent the floor required to boot Core RDK Broadband, run the required components for each profile (WAN manager for EthWAN WiFi Router; EasyMesh agent + IEEE 1905 transport for EXT EasyMesh; OneWifi, CcspPandM, telemetry common to both), and leave adequate headroom for OTA A/B partitioning and vendor value-add processes. Values are defined against, and partially validated on, the BPI-R4 (MT7988/Fillogic) reference platform; full validation and sign-off status is tracked in Section 6 (Open Items). These figures cover the Linux kernel, SoC drivers/HAL, the open-source (OSS) runtime, and Core RDK-B components together — the full software stack required to boot and run the device — not Core RDK-B in isolation; see Section 3.3 for the full scope boundary against OEM/ISP/vendor/cloud additions.

ID	Profile	Architecture (min.)	Cores	RAM (min.)	Flash (min.)	Reference Device
CRF-001	EthWAN WiFi Router	AArch64/ARMv8-A, ≥ 1.8 GHz (reference: Cortex-A73)	4 (single-cluster)	1 GB	4,096 MB / 4 GB (eMMC) — proposed; 8 GB recommended for production	BPI-R4 (MT7988/Fillogic), EthWAN build
CRF-002	EXT EasyMesh	AArch64/ARMv8-A, ≥ 1.8 GHz (reference: Cortex-A73)	4 (single-cluster)	1 GB	4,096 MB / 4 GB (eMMC) — proposed; 8 GB recommended for production	BPI-R4 (MT7988/Fillogic), EasyMesh agent-mode build

Both CRF-001 and CRF-002's flash floors are set to 4,096 MB (4 GB) minimum, with 8 GB recommended for production. Two separate storage-analysis reports (one per profile) confirm the current A/B layout already allocates ~2,172 MiB before metadata, so a 2,048 MiB part does not fit it. Both reports independently converge on the same recommendation: 2 GiB insufficient, 4 GiB technical minimum, 8 GiB recommended production capacity, 16 GiB+ optional (only for additional persistent data storage). Gateway evidence: production-style image (build rdk8-1.0.0_20260916), RootFS A 401 MiB used of 958 MiB capacity (45%, base 398.10 MiB + overlay 2.39 MiB). Extender evidence: a properly production-representative image (build rdk8-1.0.0_20260902, despite a BUILD_TYPE=dev label), RootFS A 221.44 MiB used of 957

MiB capacity (25%, base 220.96 MiB + overlay 0.50 MiB) — confirming the extender genuinely needs less flash than the gateway (no WAN stack), not more. This also resolves the earlier open item questioning CRF-002: the prior EXT figure suggesting an 8 GB need from 1.7 GiB observed usage was measured on a development-SDK-contaminated image (Go, Rust, LLVM, GCC, headers, debug packages, kernel build tree — over 1 GiB of toolchain alone); this clean measurement supersedes it. Both profiles' 4 GB/8 GB figures remain marked proposed pending formal governance approval and constrained-flash validation testing, per the same discipline applied throughout this specification.

Storage technology validation gap: the CRF table and JSON examples specify “eMMC” as the storage type, matching STG-001's supported production technologies (eMMC, UFS, or SPI-NAND). The BPI-R4 reference device does not actually use eMMC — its rootfs A/B partitions are hosted on an SD card (GPT-partitioned) for development convenience, with only the ~120 MiB boot chain on onboard SPI-NAND (see Section 3.7.1). SD is not among STG-001's supported production storage types. This means the evidence in this specification validates the software partition layout and capacity model (sizes, A/B structure, filesystem choice) but does NOT validate eMMC/UFS-specific production behavior — write endurance, bad-block handling, or power-loss safety — since none of that was exercised on the actual target storage technology. SD card is permitted for reference/development validation only; production conformance against eMMC, UFS, or SPI-NAND requires separate validation on that actual technology.

A/B banking applies to every OTA-relevant partition, not just RootFS. The reference flash layout in Section 3.7 (and the confirmed real device layout in Section 3.7.1) duplicate Bootloader (Bolt/BL2, FIP), Kernel/BootFS, and RootFS as full A-slot/B-slot pairs — each slot sized to the SAME capacity, not a partial or delta size. Concretely: if RootFS uses 500 MiB, the device needs capacity for TWO 500 MiB RootFS slots (1,000 MiB total for RootFS alone), because a safe firmware upgrade writes and verifies the complete new image into the passive slot before switching boot selection — there is no reliable A/B upgrade path that shares space between active and standby images. This is exactly why the 4 GB/8 GB flash revision above already assumes 2× RootFS (not 1×): treating a single measured RootFS size as the full device flash requirement is the single most common flash-sizing error, and this specification does not make it. The documented exception is NVRAM/persistent configuration, which real device evidence (Section 3.7.1, partition p9) shows implemented as a SINGLE SHARED partition rather than duplicated A/B — current configuration state does not need a standby copy the way an executable firmware image does.

CRF-002's CPU and RAM floors are set to 1.8 GHz and 1,024 MB / 1 GB, based on two separate EXT measurement reports. Both reports' own reference platform ran at 1.8 GHz — the prior 1.4 GHz figure was never tested; it was an architectural estimate derived by subtracting the WAN-stack workload from EthWAN's measured clock speed. Both reports also explicitly evaluated a 1 GB RAM candidate and flagged 512 MB as “not recommended — insufficient evidence,” a caution that was never retracted or acted on until now. Measured RAM usage itself (172–176 MiB used across two sessions, out of 3.83 GiB visible) is not in question and comfortably supports 1 GB; the correction is about which floor the specification states, not a change in what was actually observed. Both figures remain marked proposed pending formal governance approval and constrained-resource validation testing (loaded CPU sampling and a soak test at the stated floor), per the same discipline applied throughout this specification.

Profile convergence: EthWAN WiFi Router and EXT EasyMesh now share the same CPU architecture floor (AArch64/ARMv8-A ≥ 1.8 GHz, 4-core, single-cluster, reference Cortex-A73), the same RAM floor (1 GB), and the same flash floor (4 GB minimum / 8 GB recommended). The two profiles no longer differ in underlying hardware tier — only in WAN stack presence, measured image size (398 MiB gateway base vs. 221 MiB extender base), and EasyMesh role (controller-optional vs. agent-only, per Section 4). A single BOM/SKU sized to either profile's floor now satisfies both; this may be relevant to OEM hardware planning even though it was not a design goal of this revision — it fell out of following the evidence for each profile independently.

3.1 Sizing Notes

- CPU minimums assume a heterogeneous Arm big.LITTLE or symmetric multi-core SoC capable of concurrent packet processing (NAT/offload) and userspace RDK-B component execution; single-core SoCs are not supported. This offload capability is architecturally assumed, not confirmed by measurement — every CPU sample taken to date was idle or light-load; no loaded-forwarding test has been run that would reveal whether hardware offload is active. If a target SoC lacks hardware offload, CPU headroom needs a software-forwarding budget added, the same way RAM gets +256 MB (see the RAM bullet below).

- RAM minimums assume hardware NAT/flow offload is available on EthWAN WiFi Router; software-only forwarding paths should budget an additional 256 MB headroom.
- Flash minimums assume a dual-bank (A/B) layout for OTA, which requires roughly double the flash of an equivalent single-bank design (two full kernel + rootfs copies vs. one). Single-bank designs use less flash but do not have a standby image to fall back to, and MUST implement an alternative recovery mechanism (e.g., a dedicated recovery partition or a field-service reflash procedure) to achieve comparable OTA safety.
- EXT EasyMesh figures assume no WAN stack is loaded (WAN Interface Features are marked not applicable for this profile in the component list), reducing both RAM and flash footprint relative to EthWAN WiFi Router.

3.1.1 Storage Conformance Modes

This specification defines two explicit, mutually exclusive conformance modes for flash/OTA design. A conforming device MUST declare which mode it implements.

- **A/B Storage Mode (default):** the device implements dual-bank (A/B) partitions for bootloader, kernel/BootFS, and RootFS (Section 3.6's A/B banking principle; FLC-001–003), sized per the documented reference allocation. The 4 GB minimum / 8 GB recommended floor (CRF-001, CRF-002) applies to this mode specifically — all real BPI-R4 evidence in this specification (Section 3.7.1, Section 5.1) was measured against a device in this mode. PER-006 and FLC-001–003 assume A/B Storage Mode unless a device explicitly declares otherwise.
- **Single-Bank Recovery Mode:** the device implements a single active bootloader/kernel/RootFS (no standby A/B copy), and instead provides an independently bootable, signature-verified recovery mechanism (a dedicated recovery partition or field-service reflash path, meeting PCH-015/PCH-019's signing requirements) that does not depend on the primary, potentially-corrupted image to recover the device. The CRF-001/CRF-002 4 GB floor does NOT apply to this mode — that figure is specific to the A/B model. This specification does not currently define a complete single-bank storage-minimum calculation; a vendor choosing this mode must independently size and document their own capacity justification (single active image + recovery mechanism footprint), consistent with Section 3.3's OEM-owned scope boundary.

This is a newly formalized distinction (v23) — prior revisions described single-bank as generically “allowed with alternative recovery” without naming it as a separate conformance mode with its own capacity model. No single-bank device has been measured or evidenced anywhere in this specification; this mode remains a permitted but unvalidated design path.

3.2 Publishing Format (JSON)

These metrics are published as JSON for consumption by the docs site UI (cpokuru.github.io/CoreRDK-Broadband-Platform) and by any other tooling that needs to read or validate hardware requirements programmatically. To keep the primary spec file stable while measured data (RAM, flash, CPU) is refined independently, publishing is split across four sibling schemas per profile rather than one monolithic file:

Schema	File (per profile)	Purpose	Update Cadence
hw-compat.schema.json	<profileId>.json	CPU/RAM/flash minimums and peripheral requirements — the primary spec file.	Rare — only when minimums or peripheral requirements change.
memory-footprint.schema.json	memory-footprint/<profileId>.memory-footprint.json	Per-process RSS, answering “what consumes the RAM minimum.”	As components are profiled.
flash-layout.schema.json	flash-layout/<profileId>.flash-layout.json	Logical flash partition classes, answering “what is the flash layout.”	As real image sizes are measured.

Schema	File (per profile)	Purpose	Update Cadence
cpu-utilization.schema.json	cpu-utilization/<profileId>.cpu-utilization.json	Per-process CPU utilization, answering “what consumes the CPU minimum.”	As components are profiled.

Each profile's primary file (hw-compat.schema.json) carries a memoryFootprintRef, flashLayoutRef, and cpuUtilizationRef pointing to its three companion files, so a consumer can start at the primary file and follow the refs to whichever resource breakdown it needs. Anyone defining or updating metrics for a profile — in this document or in a future revision — should publish the corresponding JSON files so the UI component viewer can render them directly rather than parsing prose or tables.

3.2.1 Primary Schema (hw-compat.schema.json)

```
{
  "$schema": "https://cpokuru.github.io/CoreRDK-Broadband-Platform/schemas/hw-
  compat.schema.json",
  "specVersion": "1.0",
  "profileId": "string",
  "profileName": "string",
  "cpu": {
    "architecture": "string",
    "family": "string",
    "minClockGHz": 0.0,
    "minCores": 0,
    "coreTopology": "string"
  },
  "memory": {
    "minRamMB": 0
  },
  "requiredStorage": {
    "minFlashMB": 0,
    "supportedTypes": ["string"],
    "partitionLayout": "string",
    "filesystem": "string"
  },
  "referenceDevice": {
    "name": "string",
    "soc": "string",
    "buildBranch": "string"
  },
  "validationEvidence": {
    "storageType": "string",
    "productionStorageValidated": false
  },
  "peripherals": { "...": "see Section 4 peripheral schema" },
  "validationStatus": "not-started | partial | validated",
  "lastValidated": "YYYY-MM-DD",
  "memoryFootprintRef": "memory-footprint/<profileId>.memory-footprint.json",
  "flashLayoutRef": "flash-layout/<profileId>.flash-layout.json",
  "cpuUtilizationRef": "cpu-utilization/<profileId>.cpu-utilization.json"
}
```

Field	Type	Required	Description
specVersion	string	Required	Version of this JSON schema (semantic version), e.g. "1.0".
profileId	string	Required	Machine-readable profile identifier, kebab-case (e.g. "ethwan-wifi-router", "ext-easymesh").
profileName	string	Required	Human-readable profile name for display, matching Section 1.1 (e.g. "EthWAN WiFi Router").
cpu.architecture	string	Required	Instruction set / architecture family, e.g. "arm64".
cpu.family	string	Required	SoC core family, e.g. "Cortex-A73".
cpu.minClockGHz	number	Required	Minimum sustained clock speed in GHz.
cpu.minCores	integer	Required	Minimum number of CPU cores.
cpu.coreTopology	string	Optional	Core arrangement note, e.g. "single-cluster", "dual-cluster" (dual-cluster applies to big.LITTLE-class SoCs; MT7988/Fillogic 880 is single-cluster quad Cortex-A73).
memory.minRamMB	integer	Required	Minimum RAM in megabytes.
requiredStorage.minFlashMB	integer	Required	Minimum flash storage in megabytes.
requiredStorage.supportedTypes	array	Required	Flash technologies permitted in production, e.g. ["eMMC", "UFS", "SPI-NAND"]. This is the requirement, not necessarily what the reference device in validationEvidence was tested on.
requiredStorage.partitionLayout	string	Optional	OTA partition layout, e.g. "A/B", "single-bank".
requiredStorage.filesystem	string	Optional	Filesystem used on flash — "ext4", "squashfs", or "EROFS" for eMMC/UFS; "UBI/UBIFS" for SPI-NAND. MUST be compatible with requiredStorage.supportedTypes; UBI/UBIFS MUST NOT be paired with eMMC or UFS.
referenceDevice.name	string	Required	Name of the validated reference device, e.g. "BPI-R4".
referenceDevice.soc	string	Optional	SoC used on the reference device, e.g. "MediaTek MT7988 (Fillogic)".
referenceDevice.buildBranch	string	Optional	Source branch used for the validated build.
validationEvidence.storageType	string	Optional	Storage technology ACTUALLY present on the reference/validation device, e.g. "SD" — may differ from requiredStorage.supportedTypes (the production requirement).

Field	Type	Required	Description
validationEvidence.productionStorageValidated	boolean	Optional	True only when the actual storage technology used by the reference device is permitted by requiredStorage.supportedTypes and has successfully completed the applicable endurance, power-loss, OTA, recovery, filesystem, and stability validation.
validationStatus	string	Required	One of: "not-started", "partial", "validated".
lastValidated	string	Optional	ISO-8601 date (YYYY-MM-DD) of the most recent validation pass. Under validationStatus: "partial", this means "last tested" — the date some subset of the profile's values were exercised on the reference device — not a claim that every minimum in this file was confirmed at its exact stated floor. See the note below.

3.2.2 Example — EthWAN WiFi Router

```
{
  "specVersion": "1.0",
  "profileId": "ethwan-wifi-router",
  "profileName": "EthWAN WiFi Router",
  "cpu": {
    "architecture": "arm64",
    "family": "Cortex-A73",
    "minClockGHz": 1.8,
    "minCores": 4,
    "coreTopology": "single-cluster"
  },
  "memory": {
    "minRamMB": 1024
  },
  "requiredStorage": {
    "minFlashMB": 4096,
    "supportedTypes": ["eMMC", "UFS", "SPI-NAND"],
    "partitionLayout": "A/B",
    "filesystem": "ext4"
  },
  "referenceDevice": {
    "name": "BPI-R4",
    "soc": "MediaTek MT7988 (Filologic)",
    "buildBranch": "rdkb-2025q4-kirkstone"
  },
  "validationEvidence": {
    "storageType": "SD",
    "productionStorageValidated": false
  },
  "peripherals": { "...": "see Section 4" },
  "validationStatus": "partial",
}
```

```

"lastValidated": "2026-08-25",
"memoryFootprintRef": "memory-footprint/ethwan-wifi-router.memory-footprint.json",
"flashLayoutRef": "flash-layout/ethwan-wifi-router.flash-layout.json",
"cpuUtilizationRef": "cpu-utilization/ethwan-wifi-router.cpu-utilization.json"
}

```

3.2.3 Example — EXT EasyMesh

```

{
  "specVersion": "1.0",
  "profileId": "ext-easymesh",
  "profileName": "EXT EasyMesh",
  "cpu": {
    "architecture": "arm64",
    "family": "Cortex-A73",
    "minClockGHz": 1.8,
    "minCores": 4,
    "coreTopology": "single-cluster"
  },
  "memory": {
    "minRamMB": 1024
  },
  "requiredStorage": {
    "minFlashMB": 4096,
    "supportedTypes": ["eMMC", "UFS", "SPI-NAND"],
    "partitionLayout": "A/B",
    "filesystem": "ext4"
  },
  "referenceDevice": {
    "name": "BPI-R4",
    "soc": "MediaTek MT7988 (Filogic)",
    "buildBranch": "rdkb-2025q4-kirkstone"
  },
  "validationEvidence": {
    "storageType": "SD",
    "productionStorageValidated": false
  },
  "peripherals": { "...": "see Section 4" },
  "validationStatus": "partial",
  "lastValidated": "2026-08-25",
  "memoryFootprintRef": "memory-footprint/ext-easymesh.memory-footprint.json",
  "flashLayoutRef": "flash-layout/ext-easymesh.flash-layout.json",
  "cpuUtilizationRef": "cpu-utilization/ext-easymesh.cpu-utilization.json"
}

```

validationStatus should be updated as evidence accumulates: "not-started" prior to any bring-up testing, "partial" once the profile boots and runs on the reference device (its current state per Section 6), and "validated" only once the exact minimums have been confirmed stable per the methodology in Section 5 — not just the reference SKU's actual, typically higher, specs. Under "partial", lastValidated means the date something was last tested, not that every field in the file was confirmed at its exact stated value — both in-scope profiles are currently "partial" while several individual figures (e.g., EXT EasyMesh's CPU/RAM/flash floors, Section 3) remain unvalidated at their specific numbers. The three companion files (memory-footprint, flash-layout, cpu-utilization) each carry their own independent measured flag for the same reason — a profile can be validated on CPU/RAM/flash minimums while its per-process breakdown is still pending.

3.3 Scope Boundary — Core RDK-B vs. Vendor / SoC / OEM / Operator Additions

The CPU, RAM, and flash minimums in Section 3, and the measured footprint below, cover the Linux kernel, SoC drivers/HAL, the open-source (OSS) runtime, and Core RDK-B components together — the full software stack required to boot and run the device — not Core RDK-B in isolation. They are a floor, not a full production budget. SoC vendor, OEM, and operator/ISP additions on top of Core RDK-B are explicitly out of scope for this specification; sizing, validating, and qualifying hardware for those additions is the responsibility of the party adding them, not something Core RDK-B guarantees or accounts for.

- SoC-specific drivers & HAL — supplied and sized by the silicon vendor.
- OEM components (e.g., firmware upgrade tooling) — supplied and sized by the ODM/OEM.
- ISP services & configuration — defined and sized by the operator.
- Vendor apps (e.g., Cujo, speed test) — supplied and sized by the vendor providing them.
- Cloud management (e.g., OpenSync) — supplied and sized by the cloud management provider.
- Home automation services and additional protocol support — supplied and sized by whichever party adds them.

Real measured RAM/CPU/flash evidence for both in-scope profiles is in Section 5.1, gathered from actual reference-device measurement rather than illustrative figures.

The complete, current per-process RSS breakdown for the BPI-R4 reference platform — including EasyMesh controller/agent processes, which dominate RDK-B's own memory footprint — is in the memory-footprint companion JSON file referenced from Section 5.1, not reproduced here to avoid a second, smaller table drifting out of sync with the real one.

Implication for this specification: the Section 3 minimums cover the Linux kernel, SoC drivers/HAL, the open-source (OSS) runtime, and Core RDK-B components together — the full software stack required to boot and run the device — not Core RDK-B in isolation. Silicon vendors, OEMs, and operators evaluating a hardware design against this specification must add their own headroom for OEM tooling, ISP services, vendor apps, cloud management, home automation, and any additional protocols they plan to ship — SoC drivers/HAL are already included in the Section 3 floor, not an additional layer to budget for — Core RDK-B conformance to Section 3 does not imply a production device built on those minimums has room for those additions.

3.4 Extended Compute Architecture Requirements

Sections 3 and 3.1 define the CPU floor as architecture family, clock speed, and core count. The requirements below extend that floor to the fuller set of compute-architecture properties a hardware compatibility specification is expected to define — performance benchmarking, cache coherency, cryptographic and packet-processing acceleration, frequency governor behavior, thermal throttling, endianness, and userspace bitness.

ID	Item	Requirement	EthWAN WiFi Router	EXT EasyMesh
CA-001	Supported CPU architecture	ARMv8-A, AArch64 (64-bit). ARMv7-A/AArch32-only SoCs are not supported for new hardware.	Required	Required
CA-002	Minimum core count	4 physical cores. SMT/hyperthreading is not assumed or required.	Required	Required

ID	Item	Requirement	EthWAN WiFi Router	EXT EasyMesh
CA-003	Performance benchmark	Proposed target: sustained CoreMark ≥ 1500 per core at the profile's minimum clock speed (Section 3). This is informative guidance, not yet a MUST-level requirement — the test harness, sampling window, and pass/fail methodology are not yet finalized (see Section 5). CoreMark measures CPU compute throughput only; it is NOT interchangeable with, or a proxy for, real routed/NAT forwarding throughput, packet rate, or EasyMesh convergence performance — those require separate, workload-based conformance criteria under a loaded network condition, not a CoreMark score. CA-003 becomes normative once Section 5 defines a validation methodology for it.	Proposed (not yet normative)	Proposed (not yet normative)
CA-004	Hardware virtualization	Not required. No profile currently depends on a hypervisor or virtualized RDK-B instance.	Not required	Not required
CA-005	MMU and cache coherency	Hardware MMU required; cache-coherent multi-core operation required (coherent interconnect across all cores, e.g. ARM CCI/CCN-class or equivalent).	Required	Required
CA-006	Cryptographic acceleration	Hardware acceleration required for AES-128/256 and SHA-256 at minimum, to support WPA2/WPA3 key handling and TLS-based management (TR-369/USP, TR-069/CWMP) without unacceptable CPU load.	Required	Required
CA-007	Packet-processing acceleration	Hardware NAT/flow offload required to meet the RAM sizing assumption in Section 3.1 (software-only forwarding requires +256 MB headroom).	Required	Not applicable (no WAN stack)
CA-008	CPU frequency governor behavior	"ondemand" or "schedutil"-class dynamic governor required; static/performance-only governors are permitted only if the minimum clock in Section 3 is sustained continuously without thermal throttling.	Required	Required
CA-009	Thermal management	The platform MUST apply graceful, governor-level frequency throttling before reaching the SoC vendor's defined critical temperature, and MUST NOT drop sustained throughput below the Section 3 CPU floor under normal operating conditions while throttling. If temperature remains at or above the critical limit despite throttling, the platform MUST perform a controlled protective shutdown — throttling does not replace protective shutdown, and shutdown always takes precedence over maintaining throughput. Ambient/case thermal limits themselves are OEM environmental scope (Section 7.7).	Required	Required
CA-010	Endianness	Little-endian (LE) required. Big-endian SoCs are not supported.	Required	Required
CA-011	32/64-bit userspace	64-bit (AArch64) userspace required for all Core RDK-B components. 32-bit compatibility layers are not required and not recommended.	Required	Required
CA-012	AI/NPU accelerator	Not applicable for either current in-scope profile. Where a future profile requires on-device AI/ML inference at the	Not applicable	Not applicable

ID	Item	Requirement	EthWAN WiFi Router	EXT EasyMesh
	(future provision)	RDK-B layer, this item declares: accelerator architecture/vendor, performance class (TOPS or equivalent), memory interface (shared system RAM vs. dedicated accelerator memory — ties to MEM-002's RAM reservation for NPU), and runtime/driver requirement. See Section 4.6 for the corresponding physical M.2 slot provision.		

3.5 Extended Memory Requirements

minRamMB in Section 3 refers to RAM visible to and usable by Linux, not raw installed DRAM capacity. Any RAM reserved for radio firmware, an NPU/accelerator, or a secure world (TrustZone-class) is additional to, not drawn from, this floor, and must be documented per SKU by the SoC vendor or OEM under the Section 3.3 scope boundary.

ID	Item	Requirement	EthWAN WiFi Router	EXT EasyMesh
MEM-001	Minimum physical RAM	See Section 3 ("minRamMB"). This is the floor for RAM visible to Linux after firmware/NPU/Wi-Fi/secure-world reservations — see "Usable vs. installed RAM" below.	1024 MB (see Section 3)	1024 MB (see Section 3)
MEM-002	RAM reserved for firmware/NPU/Wi-Fi/secure world	Any RAM reserved outside Linux (radio firmware, NPU/accelerator, TrustZone/secure world) is additional to, not drawn from, minRamMB. Reserved amounts MUST be documented per SKU by the SoC vendor/OEM (Section 3.3) so minRamMB is verifiably the Linux-visible floor.	Required to document, vendor-owned	Required to document, vendor-owned
MEM-003	Memory visible to Linux	minRamMB in Section 3 refers specifically to RAM visible to and usable by Linux, not raw installed DRAM capacity. A device with 1 GB installed DRAM but only 850 MB visible to Linux after reservations does not meet a 1024 MB minRamMB floor.	Required interpretation	Required interpretation
MEM-004	ECC / parity	Not required for either in-scope profile. May be considered for future GW-class profiles with higher reliability requirements.	Not required	Not required
MEM-005	Swap / zram policy	zram (compressed RAM-backed swap) permitted and recommended under memory pressure; disk-backed swap is not required and not recommended given flash endurance (Section 4.3).	Recommended (zram)	Recommended (zram)
MEM-006	Peak-memory headroom	Kernel-reported MemAvailable MUST remain at or above 15% of minRamMB at steady state under the reference workload (Section 3.3 footprint) to absorb OTA staging and transient peaks. Raw MemFree is not the correct metric — Linux intentionally uses free RAM for page cache, and reclaimable page	Required	Required

ID	Item	Requirement	EthWAN WiFi Router	EXT EasyMesh
		cache, slab, and zram-backed pages count toward MemAvailable, not against this threshold.		
MEM-007	OOM behavior	Linux OOM-killer policy MUST prioritize killing non-essential/vendor processes before Core RDK-B management components (CcspPandMSSp, webpa, telemetry). Priority ordering to be defined alongside the memory-footprint companion file (Section 3.2).	Required policy, ordering pending	Required policy, ordering pending
MEM-008	Memory-leak stability criteria	After a defined warm-up period, no Core RDK-B process shall exhibit sustained monotonic RSS growth over a 72-hour soak test with representative traffic. Any growth exceeding both 10% and 10 MiB (absolute) from its post-warm-up baseline MUST be investigated and dispositioned. Final and peak MemAvailable must remain above the MEM-006 threshold throughout the test.	Required	Required
MEM-009	DMA / CMA reserved memory	Contiguous Memory Allocator regions required by Wi-Fi/Ethernet DMA are additional to minRamMB, same treatment as firmware/NPU reservations above.	Vendor-documented	Vendor-documented
MEM-010	Crash-dump memory	A reserved region for kernel crash-dump capture (e.g. kdump/ramoops-class mechanism) is recommended but not required for either in-scope profile.	Recommended	Recommended

3.6 Sizing Composition Model

Flash and RAM decompose differently and must not be sized with the same formula. Flash is additive across stored artifacts, including the bootloader, since every layer occupies permanent storage. RAM reflects only what is concurrently resident during normal operation — u-boot executes before Linux boots and is not resident afterward, so it does not belong in a runtime RAM sum.

Layer	Supplied By	In Section 3 Floor?	Notes
u-boot (bootloader)	Vendor	Flash only — not RAM	First/second-stage bootloader; "Bolt A/B" in the reference flash layout (Section 3.7); not resident at runtime, so excluded from the RAM sum.
Linux kernel & drivers, HAL	Vendor (SoC-specific)	Yes — RAM and Flash	SoC-specific kernel, drivers, and HAL — running in every measurement that produced the Section 3 floor. NOT a layer vendors budget separately on top of Section 3.
OSS (open-source runtime)	Upstream/community	Yes — RAM and Flash	Python, OpenSSL, base libc, and similar userspace packages — see the measured

Layer	Supplied By	In Section 3 Floor?	Notes
			OSS breakdown in Section 3.6/3.7.2 (~40% of measured flash for both profiles).
RDK-B	Core RDK-B	Yes — RAM and Flash	The components and reference software stack this specification is scoped to — Sections 3–5 of this document.
OEM components (e.g., FW upgrade tooling)	OEM	No — additional	Supplied and sized by the ODM/OEM; not present in any reference measurement.
Operator-specific components	Operator/vendor/cloud provider	No — additional	ISP services & configuration, vendor apps, cloud management, home automation, additional protocols — same items as Section 3.3's scope boundary.

Flash(total) = u-boot + Kernel/Drivers/HAL + OSS + RDK-B, plus OEM and Operator-specific layers added on top. RAM(min) = Kernel/Drivers/HAL + OSS + RDK-B + Reserved (firmware/NPU/secure world, Section 3.5) + Peak Transient + Headroom (MEM-006), plus OEM and Operator-specific layers added on top. Section 3's minRamMB and minFlashMB equal the first four terms of each formula combined — Kernel/Drivers/HAL, OSS, and RDK-B together, not RDK-B alone — consistent with Section 3.3. Only OEM and Operator-specific layers are genuinely additional to these numbers, and must be sized independently by whoever adds them.

3.7 Reference Flash Layout

The table below is the current reference flash partition layout, defined by RDKM. Its 53 numbered entries (rows 0–52) are allocation classes spanning the fuller RDK-B partition ecosystem across the broadband device portfolio, not 53 physical partitions on any single device — they are broader than the two in-scope profiles in this specification — it reflects the fuller RDK-B partition ecosystem across the broadband device portfolio, including GW/DOCSIS-class profiles not covered by this revision (Section 1.2). Rows shaded orange are SoC/OEM-confidential and owned by the silicon vendor or OEM per Section 3.3; rows shaded green are new or reserved-for-future-use.

#	Partition Name	Purpose	Reference Allocation
0	Bolt A	Bootloader (active)	1 MB – 2 MB
1	Bolt B	Bootloader (standby)	1 MB – 2 MB
2	Kernel A	Active kernel image	9 MB – 15 MB
3	Kernel B	Standby kernel image	9 MB – 15 MB
4	RDK-B Rootfs A	Active root filesystem	500 MB – 1 GB
5	RDK-B Rootfs B	Standby root filesystem	500 MB – 1 GB
6	Crash debug A	Crash/debug capture	20 MB – 30 MB
7	Crash debug B	Crash/debug capture	20 MB – 30 MB
8	Reserved A (remaining)	For future purpose	Remaining left flash
9	Reserved B (remaining)	For future purpose	Remaining left flash
10	DAC partition	New flash requirement for RDK-B	New requirement — planning for upcoming devices; existing leftover memory reserved for a future device-class-specific use case.
11	Reserved	Not currently allocated	—
12	MAC Address A (under Bolt)	MAC address storage	up to 200 KB

#	Partition Name	Purpose	Reference Allocation
13	MAC Address B (under Bolt)	MAC address storage	up to 200 KB
14	RDK-B NVRAM A	Persistent configuration	80 MB – 100 MB
15	RDK-B NVRAM B	Persistent configuration	80 MB – 100 MB
16	Linux Device Tree A	Device tree blob	256 KB – 512 KB
17	Linux Device Tree B	Device tree blob	256 KB – 512 KB
18	Root data A	Root data store	40 MB – 100 MB
19	Root data B	Root data store	40 MB – 100 MB
20	Bolt A NVRAM	Bootloader environment	up to 200 KB
21	Bolt B NVRAM	Bootloader environment	up to 200 KB
22	SoC/OEM specific	Confidential	up to 200 KB
23	SoC/OEM specific	Confidential	up to 200 KB
24	SoC/OEM specific	Confidential	2 MB max
25	SoC/OEM specific	Confidential	2 MB max
26	SoC/OEM specific	Confidential	9 MB – 20 MB
27	SoC/OEM specific	Confidential	9 MB – 20 MB
28	SoC/OEM specific	Confidential	2 MB max
29	SoC/OEM specific	Confidential	2 MB max
30	SoC/OEM specific	Confidential	2 MB max
31	SoC/OEM specific	Confidential	2 MB max
32	SoC/OEM specific	Confidential	256 KB – 512 KB
33	SoC/OEM specific	Confidential	256 KB – 512 KB
34	SoC/OEM specific	Confidential	1 MB – 2 MB
35	SoC/OEM specific	Confidential	1 MB – 2 MB
36	SoC/OEM specific	Confidential	1 MB – 2 MB
37	SoC/OEM specific	Confidential	1 MB – 2 MB
38	SoC/OEM specific	Confidential	1 MB – 2 MB
39	SoC/OEM specific	Confidential	7 MB – 10 MB
40	SoC/OEM specific	Confidential	7 MB – 10 MB
41	SoC/OEM specific	Confidential	up to 200 KB
42	SoC/OEM specific	Confidential	up to 200 KB
43	SoC/OEM specific	Confidential	up to 300 KB
44	SoC/OEM specific	Confidential	up to 300 KB
45	SoC/OEM specific	Confidential	250 MB – 300 MB
46	SoC/OEM specific	Confidential	250 MB – 300 MB
47	SoC/OEM specific	Confidential	up to 3 MB
48	SoC/OEM specific	Confidential	up to 3 MB
49	SoC/OEM specific	Confidential	7 MB – 10 MB
50	SoC/OEM specific	Confidential	7 MB – 10 MB
51	SoC/OEM specific	Confidential	up to 200 KB
52	SoC/OEM specific	Confidential	up to 200 KB

The DAC partition (row 10) is explicitly scoped to a specific future device class and is not applicable to either in-scope profile (EthWAN WiFi Router, EXT EasyMesh).

Reconciliation status: this flag is now resolved for BOTH in-scope profiles. CRF-001 and CRF-002 have each been revised to 4 GB minimum / 8 GB recommended, based on two separate storage-analysis reports (one per profile) that converge on

the same figures via clean, production-representative measurements. This is consistent with, and now closely matches, the scale of the reference layout above. The prior, narrower 2 GB figure for CRF-001 alone is superseded — see the corrected note above Section 3.1 for the full evidence and the retraction.

3.7.1 BPI-R4 Confirmed Partition Layout

Direct fdisk/proc evidence from the reference device confirms the actual SD-card partition scheme implemented on BPI-R4, which follows the RDKM reference layout above as its design basis, with several implementation-specific tweaks (see below). The design follows an A/B split for the four boot-critical partitions (bl2, fip, boot, rootfs); persistent/support partitions (nvram, crash, reserved, dac) sit outside that active/passive boundary and are not uniformly duplicated per slot.

Partition	GPT Label	Allocated	Filesystem	Used	Status / Purpose
p1	bl2	4 MiB	Raw boot image	N/A	BL2 A
p2	fip	2 MiB	Raw boot image	N/A	FIP A
p3	boot_a	16 MiB	VFAT	6.44 MiB (47%; FS cap 15.95 MiB)	Active BootFS
p4	rootfs_a	1,024 MiB	ext4	401 MiB	Active RootFS
p5	bl2_b	4 MiB	Raw boot image	N/A	BL2 B
p6	fip_b	2 MiB	Raw boot image	N/A	FIP B
p7	boot_b	16 MiB	VFAT	6.44 MiB (47%; FS cap 15.95 MiB)	Passive BootFS
p8	rootfs_b	1,024 MiB	ext4	0.02 MiB — effectively empty	Passive RootFS — allocated, not yet populated (upgrade target)
p9	nvram	16 MiB	ext4	4.67 MiB (37%; FS cap 13.74 MiB)	Shared NVRAM (no separate A/B)
p10	crash_a	7 MiB	ext4	0.01 MiB — effectively empty	Slot-A crash data
p11	crash_b	7 MiB	ext4	0.01 MiB — effectively empty	Slot-B crash data
p12	reserved	16 MiB	ext4	0.01 MiB — effectively empty	Reserved
p13	dac	32 MiB	ext4	0.01 MiB — effectively empty	DAC
—	Total allocated	2,172 MiB	—	—	Approximately 2.12 GiB

MAC address storage (A and B in the design intent) has no dedicated SD-card partition on this device, and — correcting an earlier inference in this specification — is not stored in the onboard SPI-NAND Factory partition either. Due to a board-level limitation, BPI-R4 does not persist MAC addresses in any non-volatile storage at all; instead, the MAC address is derived from the device serial number at boot time via a generation algorithm and used directly. MAC address storage is therefore not part of this device's flash footprint. This is a BPI-R4-specific implementation choice, not a general Core RDK-B requirement — other SoC/OEM implementations may legitimately store MAC addresses in Factory MTD (see FLC-005, Section 4.3) or elsewhere instead. “Staging” likewise has no dedicated GPT partition; since rootfs_a/rootfs_b already exist as an active/passive pair, the most likely mechanism is direct streaming into the inactive slot during OTA (one of the options STG-009 already allows) — this is an inference from the partition table, not independently confirmed.

BPI-R4's implementation is derived from the original 17-partition concept design (the simplified BL2/FIP/BootFS/RootFS × 2 + MAC/NVRAM/Crash × 2 + Reserved/Staging/DAC diagram) with several implementation-specific tweaks. This is distinct from Section 3.7's 53-entry table, which is RDKM's fuller master reference layout spanning the broader device portfolio (see the clarifying note at the start of Section 3.7) — the 53 entries are not 53 physical partitions on any single device; they are allocation classes across multiple device classes, of which the 17-partition concept is the subset relevant to this specification's two in-scope profiles.

Generic 17-Partition Concept	BPI-R4 Actual (3.7.1)	Tweak
17 partitions	13 partitions	Fewer partitions implemented
NVRAM A/B (duplicated)	Single shared NVRAM	Not duplicated — config doesn't need a standby copy
MAC Address A/B (dedicated SD partitions)	Not stored anywhere — derived from serial number at boot	Board-level limitation; computed value, not persisted data
Staging (dedicated partition)	No dedicated GPT partition	Implemented as direct-streaming into the inactive A/B slot instead (see STG-009)
Kernel A/B (separate partitions)	Folded into boot_a/boot_b (VFAT)	Kernel image + DTB combined into BootFS rather than kept separate
Bolt A/B	bl2/fip (split)	FIP broken out as its own partition pair rather than bundled under "Bolt"
Reserved A/B (duplicated)	Single reserved partition	Not duplicated
Root data A/B (dedicated 40–100 MB partitions)	Not present as a separate partition	Folded into rootfs itself
Linux Device Tree A/B (dedicated partitions)	Not present separately	DTB lives inside boot_a/boot_b alongside the kernel
Bolt NVRAM A/B	Not present separately	Likely folded into u-boot-env on SPI-NAND
DAC	Present (p13, 32 MiB)	Same, present on both

Filesystem-level detail is available for rootfs_a, the only partition measured at this depth so far:

Metric	Value
Partition allocation	1,024 MiB
ext4 usable size	958 MiB
Used	401 MiB
Available	490 MiB
Filesystem usage	45%
Inodes used	10K of 128K (prior measurement pass, not re-confirmed in the latest one)

401 MiB used + 490 MiB available = 891 MiB, not the full 1,024 MiB allocation — the 66 MiB gap between allocated (1,024 MiB) and ext4-usable (958 MiB) is standard ext4 filesystem overhead (superblock, journal, inode tables, reserved blocks), and the further 67 MiB gap between usable (958 MiB) and used+available (891 MiB) is additional ext4 reserved-block accounting. Neither gap indicates missing or corrupted data.

Minimum-flash picture: rootfs_b is currently 0.02 MiB used — effectively empty — because Slot B is an allocated upgrade target that has not yet received a firmware image, not a populated standby. This does not reduce the flash a device needs: for A/B OTA to function, Slot B must have capacity for a full image at update time, so capacity planning correctly uses rootfs_a's actual usage (401 MiB) as the per-slot target regardless of Slot B's current empty state. Total normal-filesystem usage across all 13 partitions today is approximately 420.6 MiB (rootfs_a 401 + boot_a/b 7.5 each + nvram 4.6 + crash_a/b/reserved/dac ~0.01 each from a prior measurement pass, not re-confirmed in the latest one, excluding raw BL2/FIP payloads). The device's ACTUAL current provisioning — 2,172 MiB GPT-partitioned + 120 MiB onboard SPI-NAND = 2,292 MiB (~ 2.24 GiB) — is the real flash footprint of this shipping reference build today, roughly half the current 4,096 MB (4 GB) minFlashMB floor (Section 3, CRF-001), which comfortably covers this build's actual provisioning with room to spare, consistent with the 8 GB production recommendation.

Contradiction flag: the generic reference layout above (row 10) describes the DAC partition as a “new requirement... planning for upcoming devices” reserved for a future device-class-specific use case, implying it is not yet provisioned on current-generation profiles. The BPI-R4 evidence above shows a dac partition (p13, 32 MiB) already allocated on this EthWAN WiFi Router / EasyMesh build. This needs reconciliation: either the DAC partition is provisioned as a placeholder ahead of actual use on all current-generation builds regardless of profile, or the original reference layout's scoping is inaccurate. Not resolved in this revision.

3.7.2 RootFS Base Image Composition

A directory- and subsystem-level breakdown of rootfs_a's 398.10 MiB base image (excluding the 2.39 MiB writable overlay) identifies where production-image optimization effort would have the most impact, should a smaller footprint be required for a future revision of the CRF-001 flash floor.

Directory	Used	Share of Base RootFS
/usr	314.25 MiB	84.7%
/lib	44.19 MiB	11.9%
/etc	4.67 MiB	1.3%
/bin	4.04 MiB	1.1%
/sbin	3.01 MiB	0.8%
/rdklogger	0.48 MiB	0.1%
Other	~0.29 MiB	<0.1%

/usr and /lib together account for 96.6% of the base image. The largest identifiable subsystems within them:

Subsystem	Size	Role
Python 3.10	54.7+ MiB	Runtime modules and bundled tooling
MariaDB/MySQL	~35–40 MiB	Database stack — profile applicability TBD (Section 4.2 gap)
Asterisk	29.9+ MiB	Voice stack and media — profile applicability TBD
ICU	28.7 MiB	Unicode and internationalization data
Device firmware	24.39 MiB	MediaTek, Broadcom, Realtek, and other radio/peripheral firmware
Open vSwitch	8+ MiB	Virtual switching
Perl	7.1+ MiB	Scripting runtime
ModemManager/QMI/MBIM	5–7+ MiB	Cellular management — not applicable per Section 4.1
Barton (libBartonCore)	5.12 MiB	IoT application framework
IEEE 1905	5.3+ MiB	Mesh networking transport

Optimization candidates flagged by this measurement, none yet actioned: MariaDB, Asterisk (including several MB of bundled music-on-hold audio files), ModemManager/cellular (already marked not-applicable for both in-scope profiles per Section 4.1, yet still present in the image — same build-configuration question flagged for the memory-footprint evidence in Section 5.1.1), Broadcom firmware of unconfirmed necessity on this Filogic platform, and test/debug utilities (functional test binaries, ieee1905-test-node, libasan). Trace files to their owning Yocto packages and adjust image recipes/PACKAGECONFIG rather than deleting files manually.

4. Hardware Peripheral Requirements

Each in-scope profile requires a defined set of peripheral hardware to support its connectivity role, derived from the per-profile Required/Optional/Not-Applicable feature mapping in the RDK-B Component List 2026. "Required" peripherals are mandatory for profile conformance; items marked "Optional" enable specific feature sets (e.g., Matter/Zigbee/Thread IoT) but are not required for base profile compliance. DOCSIS, xDSL, GPON/EPON, cellular, MoCA, and voice/MTA hardware are marked not applicable to both in-scope profiles per the component list and are out of scope here.

ID	Peripheral Class	EthWAN WiFi Router	EXT EasyMesh
PER-001	WAN connectivity	Required — Ethernet WAN interface (1× GbE minimum); Primary WAN + WAN profile orchestration mandatory. Secondary/backup WAN, PPPoE, GRE, and IPv6 transition mechanisms (MAP-E/T, NAT64, DS-Lite, 464XLAT) are optional.	Not applicable — no WAN interface in this profile.
PER-002	Ethernet LAN ports	Optional per component list (port management, link up/down detection, link speed management)	Wi-Fi and Ethernet backhaul both supported; at least one Ethernet

ID	Peripheral Class	EthWAN WiFi Router	EXT EasyMesh
		all marked Optional) — include if the SKU offers LAN ports.	port recommended for wired backhaul option.
PER-003	Wi-Fi radios	Required — 802.11n and 802.11ac (Wi-Fi 5) minimum standards; WPA2 required. 802.11ax (Wi-Fi 6/6E) and 802.11be (Wi-Fi 7) optional. NL80211 vendor HAL interface required.	Required — 802.11n and 802.11ac (Wi-Fi 5) minimum standards; WPA2 required. Wi-Fi AP fronthaul mandatory for extender role.
PER-004	EasyMesh role / transport	Optional — EasyMesh is not mandatory for this profile (router can run standalone AP mode).	Required — EasyMesh R6 Agent-Only role and IEEE 1905 transport are both mandatory for this profile.
PER-005	Bluetooth / IoT radio	Optional — required only if Matter/Zigbee/Thread IoT support is enabled (BLE used for Matter commissioning).	Optional — same IoT feature set applies if enabled on the extender.
PER-006	Storage interface	eMMC (ext4/squashfs/EROFS) or SPI-NAND (UBI/UBIFS) — filesystem choice must match the storage technology (see Section 4.3, STG-002); A/B partition layout recommended to support OTA firmware upgrade HAL.	eMMC (ext4/squashfs/EROFS) or SPI-NAND (UBI/UBIFS); A/B partition layout recommended for OTA.
PER-007	Media / tuner / voice hardware	Not applicable — DOCSIS, xDSL, GPON/EPON, cellular, and MTA/voice hardware are all marked not applicable for this profile.	Not applicable — no WAN, voice, or tuner hardware in scope for the extender role.
PER-008	Local WebUI / diagnostics	Optional — residential HTML/HTTPS portal typically included; UART/debug console recommended for bring-up.	Optional — typically not needed on an EasyMesh extender per component notes; UART/debug console recommended for bring-up.
PER-009	Backup/secondary WAN (cellular)	Not applicable — M.2 cellular modem, USB tethering (Android RNDIS, iPhone USB), and physical SIM card slot are all marked Not applicable per the RDK-B Component List 2026. Optional only for GW-class profiles (Section 1.2), out of scope for this revision.	Not applicable — same as EthWAN WiFi Router.

4.1 Connectivity Interface Detail

- WAN (EthWAN WiFi Router only): Ethernet WAN interface management is Required; Secondary/backup WAN, WAN failover, PPPoE, GRE, and IPv6 transition mechanisms (MAP-E/T, NAT64, DS-Lite, 464XLAT) are Optional per the component list.
- Wi-Fi: both profiles require 802.11n and 802.11ac (Wi-Fi 5) as the minimum standard. WPA2 is required; WPA3-Personal is also required (see Section 4.5). 802.11ax (Wi-Fi 6/6E) and 802.11be (Wi-Fi 7, including MLO) are Optional. EXT EasyMesh additionally requires EasyMesh R6 Agent-Only support and IEEE 1905 transport.
- IoT/BLE: Optional on both profiles — required only when Matter, Zigbee, Thread, or legacy home-security IoT support is enabled on the device (BLE is used specifically for Matter commissioning).
- Backup/secondary WAN (cellular): M.2 cellular modem, USB tethering (Android RNDIS, iPhone USB), and physical SIM card slot are all marked Not applicable for both in-scope profiles per the RDK-B Component List 2026 — see PER-009. These become Optional only for GW-class profiles (Section 1.2), which are out of scope for this revision.

4.2 Platform-Control and Recovery Hardware

Beyond connectivity peripherals, a defined set of platform-control and recovery capabilities is required for manageability, field diagnostics, and factory reset. Where the RDK-B Component List 2026 tracks these as Required/Optional software components, that status is shown below. Several items commonly expected in a hardware compatibility specification — hardware watchdog, reset button/recovery boot mode, UART console, JTAG policy, hardware RNG, unique device ID/MAC storage, and RTC — are not yet tracked as distinct items in the Component List itself; this specification defines them directly rather than leaving them unaddressed.

ID	Item	EthWAN WiFi Router	EXT EasyMesh	Source / Notes
PCH-001	Factory reset (Config Reset)	Required	Optional	Restore factory defaults via PSM and sysConfig. RDK-B Component List 2026, "Factory reset" > "Config Reset."
PCH-002	Device init & provisioning	Required	Optional	Startup scripts; sysEvent IPC + sysConfig storage. Component List, "System startup & orchestration" > "Startup Scripts."
PCH-003	Field diagnostics (remote_debugger)	Required	Required	Live field issue debugging. Component List, "Diagnostics & Testing."
PCH-004	Self-heal / recovery monitoring	Optional	Optional	Monitoring and recovery module. Component List, "Diagnostics & Testing" > "Self heal."
PCH-005	LED state management	Optional	Optional	Component List, "System startup & orchestration" > "LED Management."
PCH-006	Hardware watchdog	Required	Required	Independent hardware watchdog required, capable of forcing a reset if not serviced within a bounded interval (recommended default: 30s) by a Core RDK-B watchdog-service process. Not currently a Component-List-tracked item; this specification now defines it directly rather than leaving it untracked.
PCH-007	Reset button / recovery boot mode	A physical reset or recovery mechanism is required. Recovery boot must be reachable independently of a healthy root filesystem, for field recovery from a bad OTA. For profiles where PCH-001 factory reset is implemented, the mechanism must distinguish a factory-reset trigger (e.g., long-press) from an ordinary reboot (e.g., short-press).	Required for recovery boot access	Required for recovery boot access; factory-reset button behavior is conditional on PCH-001 being implemented
PCH-008	UART development console	Required (dev/lab units)	Required (dev/lab units)	Required on development and lab reference units for bring-up and

ID	Item	EthWAN WiFi Router	EXT EasyMesh	Source / Notes
				field debugging; MAY be physically or logically disabled on production units per the OEM's own debug-port lockdown policy (Section 7.7 boundary — the lockdown policy itself is OEM-owned, but the presence of a UART header on dev hardware is Core RDK-B's own bring-up requirement).
PCH-009	JTAG policy	Required policy, OEM-implemented	Required policy, OEM-implemented	Core RDK-B requires that a JTAG/debug-port lockdown policy exist and be documented per SKU before production release; the specific lockdown mechanism (fuse, secure-boot-gated, physically absent) is an OEM/SoC-vendor implementation choice.
PCH-010	Hardware RNG	Required	Required	A hardware random-number generator is required to seed Linux's kernel entropy pool, needed for the cryptographic acceleration requirement in Section 3.4 (WPA2/WPA3 key generation, TLS).
PCH-011	Unique device ID / MAC-address storage	Required	Required	A globally unique, stable device identifier and base/interface MAC address(es) are required. Persistence at manufacturing (e.g., in the Calibration/Factory Data partition class, Section 4.3) is one permitted implementation; deterministic regeneration from another stored unique value (e.g., device serial number) at every boot is also permitted, provided it yields a stable, collision-free result across manufactured units and survives reboot, factory reset, and OTA (see Section 3.7.1 for a real implementation of the latter approach).
PCH-012	Hardware RTC	Recommended	Recommended	A battery- or capacitor-backed real-time clock is recommended so a meaningful wall-clock time is available immediately at boot, before any network path exists.
PCH-013	WPS button	Optional	Optional	Physical WPS (Wi-Fi Protected Setup) button for simplified client onboarding. Not required by either in-scope profile; recommended for consumer-facing EthWAN WiFi Router SKUs.

ID	Item	EthWAN WiFi Router	EXT EasyMesh	Source / Notes
PCH-014	Physical status LED indicators	Recommended (Power/WAN/Wi-Fi); Optional (LAN/Cellular)	Recommended (Power/Wi-Fi); Optional (LAN)	Physical indicators for Power, WAN, LAN (per-port or aggregate), Wi-Fi, and Cellular (only relevant if PER-009 cellular backup WAN is populated). Distinct from PCH-005's software LED state management capability.
PCH-015	Secure/verified boot	Required	Required	Boot chain MUST verify each stage's signature before execution (bootloader → kernel → rootfs). See FLC-001/002/003, Section 4.3.
PCH-016	Signed firmware images	Required	Required	OTA images MUST be cryptographically signed; device MUST reject unsigned or invalid-signature images.
PCH-017	Anti-rollback protection	Required	Required	Device MUST prevent downgrade to a firmware version with known vulnerabilities below an enforced minimum version.
PCH-018	Hardware-backed key storage	Recommended	Recommended	Recommended where the SoC provides it (ties to CA-006 cryptographic acceleration) — device-unique keys protected from extraction.
PCH-019	Signed recovery image	Required	Required	Recovery/rescue boot path (PCH-007) MUST also be signature-verified, not just the primary boot chain.
PCH-020	Trusted-time recovery	Required	Required	Where no hardware RTC is present (or its value cannot be trusted, e.g. after a reset), the platform MUST establish a trusted time source via a defined secure bootstrap sequence before TLS/certificate validation or secure boot chain steps that depend on time — e.g., a signed/authenticated time source, or a bounded fallback that does not silently trust unauthenticated NTP before establishing a secure channel. NTP alone, with no bootstrap definition, does not satisfy this requirement.

Platform-control hardware (watchdog, reset/recovery, UART, JTAG, RNG, device ID/MAC storage, RTC) is defined directly in this specification (Section 4.2) rather than left as a gap. These should still be formally cross-referenced against the RDKB HAL Interface Specification's 33 status-coded interfaces (see Section 2) so the Component List and HAL spec catch up to this table rather than diverging from it.

4.3 Flash Layout — Logical Partition Classes

Beyond the aggregate minFlashMB floor in Section 3, the flash must support a defined set of logical partition classes for Core RDK-B's firmware-upgrade HAL and A/B OTA model to function. These are conceptual partition classes; Section 3.7 now provides the full authoritative reference partition table with real allocation ranges — this table is a simplified summary of that reference layout scoped to what Core RDK-B's OTA model directly requires.

ID	Partition Class	Purpose	OTA Treatment	Owner
FLC-001	Bootloader	U-Boot or equivalent first/second-stage bootloader. MUST be signature-verified as part of the secure boot chain (see PCH-015).	Platform-required. Bolt A/B (Section 3.7) form their own fallback pair, but the bootloader itself is not directly rewritten by the RDK-B OTA mechanism.	Vendor
FLC-002	Kernel (A) / (B)	Active and standby kernel images for A/B OTA fallback. MUST be signature-verified as a signed firmware image (see PCH-016).	Duplicated per slot; directly updated by OTA.	Core RDK-B
FLC-003	Rootfs (A) / (B)	Active and standby root filesystem (ext4/squashfs/EROFS on eMMC, or UBI/UBIFS on SPI-NAND) containing Core RDK-B components. MUST be signature-verified as a signed firmware image (see PCH-016).	Duplicated per slot; directly updated by OTA.	Core RDK-B
FLC-004	Persistent/Config	PSM/TR-181 persistent data store for customer-resettable configuration (Wi-Fi SSID/password, LAN settings). Cleared by factory reset; distinct from Calibration/Factory Data (FLC-005), which is never cleared by factory reset.	Platform-required; preserved across update (not directly overwritten by OTA).	Core RDK-B
FLC-005	Calibration/Factory Data	Radio calibration data, serial number; written at manufacturing. Non-resettable — NOT cleared by factory reset (see FLC-004 and STG-007). MAC address storage here is OEM/SoC-implementation-dependent — some implementations derive MAC from the stored serial number at boot rather than storing it separately (see Section 3.7.1).	Platform-required; preserved across update and factory reset; never modified by OTA.	Vendor

Per-partition sizing for each in-scope profile is tracked in its companion flash-layout.schema.json file (see Section 3.2); it should now be populated from the reference allocations in Section 3.7 rather than left at "TBD," scoped to only the rows applicable to each in-scope profile (i.e., excluding the DAC partition and any SoC/OEM-confidential rows the vendor has not disclosed). CPU utilization by process is tracked the same way, in each profile's companion cpu-utilization.schema.json file; unlike the memory footprint in Section 3.3, no CPU utilization measurement has been taken for either in-scope profile yet.

Beyond partition classes, the flash and persistent-storage subsystem as a whole must meet the following requirements:

Before the requirements below: the flash technologies relevant to this specification differ in how the storage controller and wear-leveling logic are implemented — this affects filesystem compatibility (STG-002) and OTA reliability.

Technology	What It Is	Managed By
eMMC	NAND flash with an integrated controller presenting a standard block-device interface (like a small SSD).	Onboard controller handles wear-leveling/bad-block internally — OS sees a normal block device.
UFS	Successor to eMMC — same managed-NAND/block-interface model, higher throughput via a different physical/electrical interface.	Same as eMMC — onboard controller, block-device interface.
SPI-NAND	Raw NAND accessed over a simple SPI bus, with no onboard wear-leveling controller.	Wear-leveling and bad-block management handled by software (Linux MTD + UBI layer).
NOR flash	A distinct flash memory architecture from NAND — slower, byte-addressable, small capacity, capable of execute-in-place (XIP).	Usually no filesystem; used for raw boot code that must run before the full flash subsystem initializes.

Valid storage-technology / filesystem combinations (STG-002):

Storage Type	Valid Filesystem(s)		Why	
eMMC	ext4, squashfs, EROFS		Block device — standard Linux filesystems apply directly.	
UFS	ext4, squashfs, EROFS		Same as eMMC — block device.	
SPI-NAND	UBI/UBIFS only		Requires an MTD/NAND target; UBI provides the wear-leveling/bad-block layer software-side. MUST NOT be paired with eMMC or UFS — both already have onboard wear-leveling.	
ID	Item	Requirement	EthWAN WiFi Router	EXT EasyMesh
STG-001	Supported flash technologies	eMMC, UFS, or SPI-NAND. Raw NAND without a managed FTL/UBI layer is not supported; NOR flash is permitted only for the bootloader partition class.	Required (eMMC/SPI-NAND)	Required (eMMC/SPI-NAND)
STG-002	Valid filesystem combinations	UBI/UBIFS for SPI-NAND (raw NAND behind an MTD target). ext4, squashfs, or EROFS for eMMC (block device). UBIFS requires an MTD/NAND target and MUST NOT be paired with eMMC — "UBI/UBIFS over eMMC" is not a supported combination. Filesystem choice MUST support the A/B partition layout in Section 4.3.	Required	Required
STG-003	Crash and log storage	A persistent (survives reboot) log/crash partition is required, separate from the Rootfs partitions, sized to retain at least the most recent boot cycle's crash data.	Required	Required
STG-004	Flash write endurance	Minimum rated endurance sufficient for the expected OTA cadence and persistent-config write pattern over	Required (rating TBD by	Required (rating TBD by component choice)

Storage Type	Valid Filesystem(s)		Why	
		the product's expected service life; exact P/E cycle rating is a component-selection decision made against this requirement, not a fixed number in this specification.	component choice)	
STG-005	Bad-block management	Bad-block management and wear leveling MUST be provided through the Linux MTD/UBI stack or an equivalent validated software layer for raw SPI-NAND, which has no internal managed FTL; not applicable for eMMC (handled internally by the eMMC controller).	Required where applicable	Required where applicable
STG-006	Power-loss protection	Filesystem and partition writes (especially Persistent/Config, Section 4.3) MUST tolerate an unexpected power loss without corrupting the active, bootable partition set.	Required	Required
STG-007	Factory-reset behavior	MUST restore Persistent/Config (FLC-004) — customer-resettable configuration such as Wi-Fi SSID/password and LAN settings — to defaults (PSM and sysConfig, per Section 4.2) without requiring a full reflash. MUST NOT touch Calibration/Factory Data (FLC-005), which is non-resettable identity/calibration data and is never cleared by factory reset. This requirement applies only when factory-reset (Config Reset, PCH-001) capability is implemented on the profile; it does not independently mandate factory-reset capability where PCH-001 does not require it.	Required	Conditional — required only if PCH-001 factory reset is implemented (not required standalone; PCH-001 is Optional for this profile)
STG-008	Secure erase	Required for decommissioning, RMA, and ownership transfer — broadband gateways store Wi-Fi credentials, operator credentials, certificates/keys, and subscriber configuration. Acceptable mechanisms include cryptographic erase (key destruction), secure overwrite where supported, or filesystem/flash-specific sanitization. Manufacturing calibration and immutable device identity (Section 4.2, PCH-011) are preserved separately under their own policy and are not in scope of this erase requirement.	Required	Required
STG-009	OTA staging mechanism	The OTA update mechanism MUST provide an explicit staging path for an incoming image — e.g., a dedicated staging partition, direct streaming into	Required	Required

Storage Type	Valid Filesystem(s)		Why	
		the inactive (standby) A/B slot, or another defined mechanism. Unallocated flash space alone does not provide staging capability unless the partition/update design explicitly uses it; where a dedicated staging partition is used, it MUST be sized to hold at least one full update image.		
STG-010	Upgrade image size limit	OTA image size MUST fit within the standby (B) Kernel + Rootfs partition allocation defined per-profile in the companion flash-layout.schema.json file (Section 3.2).	Required	Required

4.4 Network Hardware Requirements

Beyond the connectivity peripherals in the top-level table above, the following network-hardware requirements apply to the Ethernet and switching subsystem for each in-scope profile.

ID	Item	Requirement	EthWAN WiFi Router	EXT EasyMesh
NET-001	Ethernet MAC and PHY	Gigabit-capable MAC/PHY required on all Ethernet ports (WAN and LAN).	Required	Required, where an Ethernet port is present (EXT profile permits Wi-Fi-only backhaul)
NET-002	Required port speeds	10/100/1000 Mbps auto-negotiating; 2.5GbE recommended on WAN for EthWAN WiFi Router.	Required (GbE min.)	Required (GbE min.), where LAN ports present
NET-003	Auto-negotiation	IEEE 802.3 auto-negotiation required on all Ethernet ports.	Required	Required, where an Ethernet port is present
NET-004	Full/half duplex	Full-duplex required; half-duplex fallback permitted only for auto-negotiation compatibility with legacy link partners.	Required (full-duplex)	Required, where an Ethernet port is present
NET-005	VLAN support	802.1Q VLAN tagging required on the WAN interface to support operator tagged-WAN deployments (see Section 4.1).	Required	Not applicable (no WAN interface)
NET-006	Jumbo frame policy	Not required. Standard 1500-byte MTU is the baseline; jumbo frame support is an optional vendor value-add.	Not required	Not required
NET-007	WAN/LAN port identification	Physical WAN port MUST be distinguishable from LAN ports (labeling or distinct connector/LED) for field diagnostics.	Required	Not applicable
NET-008	Switch/bridge capacity	Onboard switch MUST support the minimum LAN port count offered by the SKU with line-rate forwarding between all ports.	Required, where LAN switch present	Required, where LAN switch present

ID	Item	Requirement	EthWAN WiFi Router	EXT EasyMesh
NET-009	Hardware NAT/flow offload	See Compute Architecture, "Packet-processing acceleration" above — required for EthWAN WiFi Router.	Required	Not applicable
NET-010	Maximum MAC-table size	Minimum 1024 entries to support typical residential/small-business client counts plus EasyMesh topology overhead.	Required (≥1024)	Required (≥1024), where an Ethernet port is present
NET-011	Maximum connection count	Minimum 32,000 concurrent NAT/contrack sessions.	Required (≥32,000)	Not applicable
NET-012	SFP/SFP+ support	Not required for either in-scope profile; relevant to a future GW/DOCSIS-class profile.	Not applicable	Not applicable
NET-013	GPON/XGS-PON, DOCSIS, DSL interfaces	Not applicable — both in-scope profiles use Ethernet WAN (EthWAN WiFi Router) or no WAN (EXT EasyMesh). See Section 1.2 for profiles where these apply.	Not applicable	Not applicable
NET-014	Cellular modem interfaces	Not applicable per Section 4.1 cellular peripheral entry — marked not-applicable in the RDK-B Component List 2026 for both in-scope profiles.	Not applicable	Not applicable
NET-015	USB tethering / PCIe/M.2 modem integration	Not applicable for either in-scope profile.	Not applicable	Not applicable

4.5 Wi-Fi Hardware Requirements

Beyond the top-level Wi-Fi peripheral entry, the following requirements define the full radio hardware and regulatory-adjacent capability set. Regional operating-mode requirements (6 GHz, DFS, country-code enforcement) are defined at the platform-behavior level here and at the per-region certification-evidence level in Section 7.

Known validation gap: TR-181 statistics for MLO-associated clients are incomplete on the reference platform — Stats.RetransCount stuck at 0, SignalStrength/RSSI reads 0 for MLO clients, OperatingStandard empty for 802.11be clients. This MUST be resolved before MLO can be marked validated rather than optional (see Section 6).

ID	Item	Requirement	EthWAN WiFi Router	EXT EasyMesh
WIFI-001	Supported bands	2.4 GHz and 5 GHz required; 6 GHz optional (see Section 7.3 for regional operating-mode requirements when present).	Required (2.4/5 GHz); 6 GHz optional	Required (2.4/5 GHz); 6 GHz optional
WIFI-002	Required 802.11 generations	802.11n and 802.11ac (Wi-Fi 5) minimum, per Section 4. 802.11ax (Wi-Fi 6/6E) and 802.11be (Wi-Fi 7, incl. MLO) optional.	Required (n/ac min.)	Required (n/ac min.)
WIFI-003	Channel widths	20/40/80 MHz required at 5 GHz minimum; 160 MHz required only where 802.11ax/be is present.	Required	Required
WIFI-004	Spatial streams	Minimum 2x2 MIMO on each supported band.	Required (≥2x2)	Required (≥2x2)

ID	Item	Requirement	EthWAN WiFi Router	EXT EasyMesh
WIFI-005	Radio-chain and antenna configuration	Antenna configuration MUST match the radio chain count; internal or external antennas both permitted. Exact design is OEM-owned (Section 7.7 mechanical exclusion) but MUST be documented per SKU for regulatory evidence (Section 7.4).	Required to document	Required to document
WIFI-006	DFS capability	Required wherever 5 GHz DFS channels are enabled for a region (see Section 7.2 — DFS detection MUST operate below the application layer).	Required, region-dependent	Required, region-dependent
WIFI-007	Transmit-power controls	Per-channel, per-band TX power control required, accounting for antenna gain (Section 7.2).	Required	Required
WIFI-008	Country-code enforcement	Required — see Section 7.2 regional configuration requirements; world-domain/unrestricted operation is prohibited on production devices.	Required	Required
WIFI-009	Regulatory database	Platform MUST ship with an up-to-date, signed wireless regulatory database appropriate to the kernel/driver stack in use, covering all regions in Section 7.6, and MUST support updating that database independent of a full firmware upgrade where the platform's driver stack allows it.	Required	Required
WIFI-010	Antenna gain	MUST be accounted for in transmit-power calculations (see above); exact gain figures are part of the per-SKU regional certification evidence (Section 7.4), not this specification.	Required to account for	Required to account for
WIFI-011	Indoor/outdoor restrictions	MUST be enforceable per band/channel per region, particularly for 6 GHz LPI (Section 7.3).	Required, region-dependent	Required, region-dependent
WIFI-012	Concurrent-radio combinations	Simultaneous 2.4/5 GHz operation required; simultaneous tri-band (2.4/5/6 GHz) required only where 6 GHz is present.	Required	Required
WIFI-013	MLO capability	Optional — Multi-Link Operation (802.11be/Wi-Fi 7), allowing simultaneous multi-band links for a single client association. Where implemented: minimum 2 simultaneous links REQUIRED; EHT-MCS up to at least MCS 12 REQUIRED where MLO is present (confirmed achievable on MT7996 tri-band MLD reference hardware); real-time load balancing across links REQUIRED where MLO is present.	Optional	Optional
WIFI-014	EasyMesh role	See Section 4 — Not required (EthWAN WiFi Router) / Required, Agent-Only (EXT EasyMesh).	Optional (per Section 4)	Required, Agent-Only (per Section 4)
WIFI-015	Radio calibration	Per-radio calibration data required at manufacturing; see Calibration/Factory Data partition class, Section 4.3.	Required	Required

ID	Item	Requirement	EthWAN WiFi Router	EXT EasyMesh
WIFI-016	WPA2/WPA3 capabilities	WPA2 required (Section 4). WPA3-Personal REQUIRED in addition to WPA2 for any profile shipping after this revision; WPA3-only mode optional. This corrects the Section 4 peripheral table, which previously listed WPA2 only.	Required (WPA2 + WPA3)	Required (WPA2 + WPA3)
WIFI-017	Protected Management Frames (PMF)	Required when WPA3 is enabled (mandatory per Wi-Fi Alliance WPA3 certification); recommended even in WPA2-only configurations.	Required with WPA3	Required with WPA3
WIFI-018	Hardware key storage	Recommended where the SoC provides it (ties to Cryptographic acceleration, Compute Architecture above); not required for either in-scope profile.	Recommended	Recommended

4.6 Expansion and Module Slots

Beyond onboard connectivity, the following expansion capabilities are defined — a general-purpose USB port (distinct from NET-015's cellular-specific USB tethering), and PCIe/M.2 slots for future accelerator and radio modules.

General-purpose USB port: Optional — for external storage or peripheral use, distinct from NET-015's USB tethering (specifically for cellular modem integration). Not required by either in-scope profile.

PCIe/M.2 expansion slots:

Slot Type	Purpose	Status (Both Profiles)
M.2 (AI/NPU accelerator)	Future NPU/AI accelerator module — ties to CA-012.	Not applicable (reserved for future profiles)
M.2/PCIe (Wi-Fi card)	Discrete Wi-Fi radio module, alternative to SoC-integrated Wi-Fi.	Not applicable
M.2 (cellular card)	Cellular modem module — ties to PER-009.	Not applicable
M.2/PCIe (IoT radio)	Discrete Zigbee/Thread/Matter radio module.	Optional — ties to existing PER-005 IoT/BLE
mSATA	Storage expansion beyond onboard flash.	Not applicable

4.7 Known Gaps to Track Against Peripheral Requirements

- The component list marks Ethernet LAN port management as Optional for EthWAN WiFi Router; this should be reconciled with actual reference SKUs, most of which ship with physical LAN ports.
- TR-181 statistics gaps observed on Wi-Fi 7 MLO clients (Stats.RetransCount, SignalStrength/RSSI, OperatingStandard) should be resolved before Wi-Fi 7/MLO-capable radios are marked as a validated (rather than optional) peripheral choice.
- Secure boot / hardware root-of-trust is now defined directly as PCH-015 through PCH-019 (Section 4.2), but is not yet tracked as a Required/Optional item in the RDK-B Component List itself — that reconciliation is still open.

5. Reference Device Validation Methodology

Compatibility values in Sections 3 and 4 are validated using the BPI-R4 (MT7988/Fillogic) platform as the primary reference device, built on Yocto Kirkstone 4.0 / Kernel 6.6 (branch rdkb-2025q4-kirkstone). Validation activities include:

- Functional sanity suite (20 test cases) covering boot, WAN establishment, and Wi-Fi bring-up — currently passing on the EthWAN WiFi Router profile build.
- EasyMesh R6 multi-node validation (controller + extender) against a 5-node CI-mandated topology; current lab is a 1-hop structural subset covering 19 of 35 CI test cases, with 16 test cases blocked pending expanded lab topology.
- Wi-Fi 7 MLO validation (55-test-case plan, CDRouter 16.x methodology) exercising the peripheral and TR-181 boundary defined in Section 4.
- HAL interface cross-reference against the RDKB HAL Interface Specification (33 interfaces) to confirm every required peripheral class in Section 4 has a corresponding, status-coded HAL binding.

Full CPU/memory/flash floor validation (i.e., confirming the profile boots and remains stable at the exact minimums in Section 3, not just on the reference SKU's actual specs) has not yet been executed and is tracked as an open item in Section 6.

5.1 Measured Evidence

Controlled runtime measurements were captured on the BPI-R4 reference device for both in-scope profiles — an EthWAN WiFi Router build running combined EasyMesh Controller/Agent roles, and a separate Fillogic tri-band EasyMesh extender build. Both measurement reports explicitly distinguish confirmed reference-platform capability from a validated minimum; neither validates the Section 3 floors, and both surface a real discrepancy worth tracking rather than silently resolving.

5.1.1 EthWAN WiFi Router

Resource	Stated Floor (Section 3)	Measured Evidence	Assessment
CPU	AArch64/ARMv8-A ≥ 1.8 GHz (CRF-001)	5.56% aggregate CPU over a 10-second sample at 1.8 GHz, governor=performance, all 4 cores online; traffic/client load not recorded.	Confirms reference capability at the stated clock; sample too short and uncontrolled to establish a minimum (Section 6).
RAM	1024 MB (CRF-001)	3925 MiB Linux-visible; 573 MiB used; 3297 MiB MemAvailable, captured ~7 minutes after boot on a development image (netdata, MariaDB, Ollama, Asterisk, ModemManager, SNMP, and other non-profile services active).	573 MiB used on a development image already leaves limited headroom under the 1024 MB floor once non-profile services are accounted for; not yet validated by constrained-RAM testing on a production image (Section 6).
Flash	4,096 MB / 4 GB (CRF-001)	Production-style image measured directly: RootFS A 401 MiB used of 958 MiB capacity (45%), base 398.10 MiB + overlay 2.39 MiB. Total device provisioning: 2,172 MiB GPT-partitioned + 120 MiB onboard SPI-NAND = 2,292 MiB (≈2.24 GiB).	Comfortably under the 4 GB floor — confirmed by dedicated storage-analysis evidence (Section 3.7).

5.1.2 EXT EasyMesh

Resource	Stated Floor (Section 3)	Measured Evidence	Assessment
CPU	AArch64/ARMv8-A ≥ 1.8 GHz (CRF-002)	2.47% aggregate CPU over a valid 60-second sample at 1.8 GHz, governor=performance; idle/light-load, shortly after boot.	Confirms reference capability at 1.8 GHz — the floor itself is now set to 1.8 GHz, matching what both dedicated EXT reports actually measured and evaluated; the prior 1.4 GHz figure was an unvalidated architectural estimate, now retired.
RAM	1,024 MB / 1 GB (CRF-002)	3.83 GiB Linux-visible; used memory has grown across three sessions: 172 MiB → 176 MiB → 582.0 MiB (fresh live-device capture, 3.26 GiB MemAvailable). The growth is dominated by onewifi_em_agent: 20.24 MiB (fuller 140-process capture) → 363.71 MiB (fresh capture), an ~18x increase and now the extender's single largest process by a wide margin (nearly 7x OneWifi's own 54.0 MiB). This is a two-point comparison (no capture exists between these sessions), but the magnitude is larger than the equivalent growth measured on the gateway's onewifi_em_ctrl/onewifi_em_agent — the strongest signal yet toward a genuine leak in the shared EasyMesh agent component.	Two earlier EXT measurement reports (172 MiB and 176 MiB used, respectively) both explicitly flag 512 MB as “not recommended — insufficient evidence” and both treat 1 GB as the real candidate floor, adopted as the current requirement. The fresh 582.0 MiB capture is still under that 1 GB floor but leaves materially less headroom than earlier sessions suggested, and the onewifi_em_agent growth pattern above means this floor should not be treated as settled until the 72-hour soak test (MEM-008) has actually been run — running it is now a priority, not a background item.
Flash	4,096 MB / 4 GB (CRF-002)	Clean production-representative image measured directly: RootFS A 221.44 MiB used of 957 MiB capacity (25%), base 220.96 MiB + overlay 0.50 MiB — confirming the extender needs less flash than the gateway (no WAN stack). An earlier measurement on a dev-SDK-contaminated image showed 1.7 GiB; that figure is superseded by this clean result.	Comfortably under the 4 GB floor — confirmed by two separate storage-analysis reports (Section 3.7).

This evidence has been used to populate the memory-footprint and cpu-utilization companion files (Section 3.2) with real per-process RSS and aggregate CPU data for both profiles, replacing the prior empty/"TBD" placeholders — each file's measured flag and notes make clear which figures are directly measured versus still outstanding (e.g., no per-process CPU% breakdown was captured, only aggregate utilization). Neither report's findings have been used to change the Section 3 minimums themselves; that remains a governance decision tracked in Section 6.

6. Open Items and Risks

- Flash floors (4 GB minimum / 8 GB recommended, both profiles) are well-evidenced by two separate storage-analysis reports. CPU and RAM floors remain architectural judgment derived from the reference device's actual configuration — neither has been independently validated by under-provisioning a reference build to the stated floor and running a constrained-resource soak test.

- EasyMesh CI topology gap (16 of 35 test cases blocked) limits how much of the EXT EasyMesh peripheral/connectivity matrix in Section 4 can be marked fully validated versus "defined."
- Five profiles from the RDK-B Component List 2026 (Modem/ONU, GW, GW OpenSync, GW EasyMesh, EXT OpenSync) are out of scope for this revision; a follow-up revision is needed to extend CPU/memory/flash and peripheral tables to those profiles, particularly GW EasyMesh and Modem/ONU which introduce DOCSIS/xDSL/GPON front-end hardware not covered here.
- The Ethernet LAN port Optional/Required status for EthWAN WiFi Router (Section 4.7) should be reconciled against actual reference SKUs before this baseline is finalized.
- CPU-utilization companion files (Section 3.2, Section 4.3) now have real per-process measured data for both in-scope profiles (idle/light-load condition only). A loaded-condition sample (EasyMesh formed, active Wi-Fi clients, sustained WAN throughput) has not yet been collected for either profile, and the profiles' assumed hardware NAT/flow offload capability (CA-007) remains unconfirmed by measurement either way; loaded-condition CPU profiling should be prioritized as a follow-up pass.
- Section 3.3's Core RDK-B vs. OEM/operator scope boundary correctly includes SoC drivers/HAL, OSS, and the kernel in the Section 3 floor. This corrected definition should be reviewed by governance to confirm it matches how the spec is meant to be enforced during hardware qualification — e.g., whether operator addenda documenting their own OEM/ISP footprint become a formal, tracked artifact alongside this specification.
- Platform-control hardware (Section 4.2) defines watchdog, recovery boot mode, UART, JTAG, RNG, device ID/MAC storage, RTC, secure boot, and signed-firmware requirements directly; these still need formal cross-reference against the HAL Interface Specification's status-coded interfaces to confirm HAL bindings exist for each.
- The full A–F requirement tables (Sections 3.4, 3.5, 4.3, 4.4, 4.5) have not yet been validated against reference hardware or exercised by the Section 5 test suite — they represent the specification's stated requirements, not confirmed measurements. Performance-benchmark methodology (Section 3.4) in particular still needs to be finalized.
- Regional applicability (Section 7) currently defines required platform behaviors and a starter JSON schema only; no region has been profiled, pre-compliance tested, or certified for either in-scope profile. This should not be read as an assertion that either profile is deployable in any region today.
- The Sizing Composition Model (Section 3.6) now has a measured OSS/OEM/kernel/RDK-B breakdown for both in-scope profiles' actual base images (Section 3.7.2), but this has not yet been cross-validated against a full per-layer memory accounting on the reference device — only the flash-side composition has been measured this thoroughly.
- EthWAN WiFi Router's measured RAM usage has grown across three snapshots: 573–631 MiB (first two, development image with netdata/MariaDB/Ollama present) → 905.6 MiB (third, production-style image without that dev tooling, per Section 5.1.1). The growth tracks two specific processes — onewifi_em_ctrl (118.7 → 236.9 → 271.9 MiB) and onewifi_em_agent (21.0 → 64.2 → 215.4 MiB, now exceeding OneWifi's own footprint) — across all three captures, a pattern materially stronger toward a genuine leak than earlier two-point data suggested. The earlier "287–330 MiB, comfortably under the 1024 MB floor" estimate no longer holds at the third capture's growth level and should not be relied on until the pattern is understood. This has not been confirmed by an actual constrained-RAM boot test at the stated floor, and the formal 72-hour soak test (MEM-008) that would distinguish leak from legitimate growth has still not been run — given this pattern, running it should be prioritized.
- EXT EasyMesh's measured RAM usage has grown even more sharply than the gateway's: 172–176 MiB (earlier sessions) → 582.0 MiB (fresh live-device capture), driven almost entirely by onewifi_em_agent growing from 20.24 MiB to 363.71 MiB (~18x) — the largest single-process growth factor observed on either profile, and now nearly 7x the size of OneWifi itself on this device. This is a two-point comparison, not three like the gateway's onewifi_em_ctrl/onewifi_em_agent history, so the growth curve shape is unconfirmed, but the magnitude alone makes this the strongest evidence yet that the shared EasyMesh agent component (present on both profiles) may have a genuine leak rather than bounded mesh-state growth. Still under the 1 GB floor at 582.0 MiB, but with materially less headroom than earlier sessions indicated. The 72-hour soak test (MEM-008) should be run against this specific component as a priority.

- 46 total failed systemd units were observed across both profiles' reference builds (21 on EthWAN WiFi Router, 25 on EXT EasyMesh) — including gateway-only services appearing on the extender-only build. Per the measurement reports' own qualification rule, a service may be dispositioned as expected/optional but must not remain unexplained; neither build can be called functionally validated until each failed unit is mapped to Required/Optional/Not-Applicable for its profile.

7. Regional Applicability

RDK-B is deployed by operators across North America, Europe, and Asia-Pacific, so hardware validated against Sections 3–4 of this specification is not automatically deployable everywhere. This section separates what Core RDK-B itself must support (the platform capability layer) from what is genuinely a certification/product-program responsibility (the regulatory evidence layer) — the same scope boundary Section 3.3 already applies to SoC/OEM/operator additions, extended to regional regulatory requirements.

7.1 Scope Boundary

Core RDK-B is responsible for defining and implementing the platform mechanisms that make correct regional operation possible. The OEM or operator product program is responsible for obtaining and maintaining the actual regulatory certifications (FCC, ISDE, RED/CE, MTCTE, ANATEL, KC, RCM, and others) for a specific hardware SKU in a specific region — Core RDK-B conformance never substitutes for that certification, and this specification does not attempt to enumerate or track country-specific regulatory clauses, which change on a timeline independent of this document.

7.2 Required Platform Behaviors

Regardless of target region, a Core RDK-B device must support the following mechanisms so that correct regional behavior is achievable once certification work is done:

- A regulatory region configuration that controls radio bands, channels, bandwidth, transmit power, DFS behavior, antenna limitations, and indoor/outdoor operation.
- Region configuration protected against unauthorized change, with factory-programmed country/sales-region data supported.
- Wi-Fi must not start in an unrestricted "world domain" mode on a production device; unsupported channels must remain disabled.
- DFS detection must operate below the application layer, and transmit power controls must account for antenna gain.
- A region change must correctly restart or reconfigure the affected radios, and calibration data must match the installed radio and antenna design.
- A single global firmware image may be used only when regional behavior is safely selected at provisioning or manufacturing time — never left to runtime guesswork.
- Regional configuration must survive OTA upgrade and factory reset, consistent with the persistent/config partition class in Section 4.3.

7.3 6 GHz Regional Operating Modes

"Wi-Fi 6E/7 supported" is not sufficient as a regional requirement. 6 GHz availability, power limits, and operating mode (Low Power Indoor, Very Low Power where permitted, Standard Power where permitted with AFC, indoor-only enforcement) vary by region, and a platform may support 6 GHz in one region while being unable to operate that

band — or requiring a different configuration — in another. The regional JSON schema in Section 7.5 captures this per region via `radios.wifi6GHz.operatingModes` rather than a single supported flag.

7.4 Regional Compliance Status Model

Regional readiness is tracked separately from the CPU/RAM/flash `validationStatus` already used in Section 3.2 — a profile can be "validated" on hardware minimums while having no regional certification evidence at all. The status ladder below should be used per profile, per region:

Status	Meaning
Hardware defined	Hardware requirements documented (Sections 3–4 of this specification).
Software compatible	Core RDK-B boots and functions on the hardware.
Performance validated	Meets load and stability targets (Section 5 methodology).
Regional configuration validated	Correct regional radio/power behavior confirmed for the target region.
Pre-compliance passed	Lab pre-scan completed ahead of formal certification submission.
Regulatory certified	Required authority/certification approval obtained (FCC, CE, ISSED, MTCTE, ANATEL, etc.).
Production qualified	Exact BOM, antenna, enclosure, and firmware combination approved for shipment.

A region must never be marked "Regulatory certified" or "Production qualified" without an actual certificate or test-report reference in the compliance array below — this mirrors the validationStatus discipline already established in Section 3.2.

7.5 Regional Publishing Format (JSON)

Regional data is published as a further companion file per profile per region — `regional/<profileId>.<regionId>.json` — kept separate from the primary hw-compat file for the same reason as the other three companion files: regional/certification status changes on its own timeline and should never require revising or re-versioning this specification.

```
{
  "profileId": "string",
  "regionId": "string",
  "regulatoryDomain": "string",
  "radios": {
    "wifi24GHz": { "supported": true },
    "wifi5GHz": { "supported": true, "dfsRequired": true },
    "wifi6GHz": { "supported": true, "operatingModes": ["LPI", "VLP", "SP"] }
  },
  "power": {
    "adapterInput": "string",
    "plugType": ["string"]
  },
  "compliance": [
    { "scheme": "string", "status": "not-tested | pre-compliance | certified",
    "certificateId": null }
  ]
}
```

Example — EthWAN WiFi Router, United States:

```
{
  "profileId": "ethwan-wifi-router",
  "regionId": "us",
  "regulatoryDomain": "FCC",
  "radios": {
    "wifi24GHz": { "supported": true },
    "wifi5GHz": { "supported": true, "dfsRequired": true },
    "wifi6GHz": { "supported": true, "operatingModes": ["LPI"] }
  },
  "power": {
    "adapterInput": "100-240 V AC, 50/60 Hz",
    "plugType": ["NEMA-1-15", "NEMA-5-15"]
  },
  "compliance": [
    { "scheme": "FCC", "status": "not-tested", "certificateId": null }
  ]
}
```

7.6 Region Coverage

The following regions are anticipated based on RDK-B's operator footprint. None has been profiled yet for either in-scope profile; this table is a tracking placeholder, not a claim of coverage.

Region	Primary Regulatory Scheme(s)	Status
North America (US)	FCC (Part 15, equipment authorization)	Not started
Canada	ISED, ICES	Not started
European Union / EEA	RED, CE marking, RoHS, REACH, WEEE	Not started
United Kingdom	UK Radio Equipment Regulations, UKCA, PSTI	Not started
India	TEC/MTCTE, WPC ETA, BIS	Not started
Japan	Radio Law / TELEC, JATE, VCCI, PSE	Not started
South Korea	KC certification	Not started
Australia / New Zealand	RCM, ACMA	Not started
Brazil	ANATEL	Not started
Other operator-selected markets	Per operator requirement	Not started

7.7 Explicitly Out of Scope

The following are real hardware-compatibility concerns but are OEM/product-certification-program responsibilities, not Core RDK-B specification content — consistent with Section 3.3's boundary:

- Mechanical compatibility (enclosure/PCB dimensions, mounting points, antenna placement, ventilation, heatsink requirements).
- Environmental and reliability testing (thermal cycling, vibration, mechanical shock, drop testing, ESD/surge/radiated/conducted immunity, MTBF, IEC 60068 / IEC 61000-4-x test families).
- Power and electrical detail beyond the regional adapter/plug fields already in the Section 7.5 schema (inrush current, brownout behavior, PoE, adapter efficiency, standby power).
- Manufacturing and lifecycle management (component traceability, second-source validation, obsolescence policy, RMA diagnostics, firmware support period commitments).

These may be worth an OEM to record in their own product-certification documentation, but including test-procedure-level detail for them here would mean this specification is asserting compliance responsibility that Core RDK-B cannot actually validate or own — the same reasoning Section 3.3 applies to SoC drivers, OEM tooling, and ISP services.

8. Glossary

- RDK-B — Reference Design Kit for Broadband.
- EasyMesh — Wi-Fi Alliance multi-AP mesh networking standard; R6 refers to the current certification release.
- GW / EXT — Gateway and Extender device roles within an EasyMesh network.
- MLO — Multi-Link Operation, a Wi-Fi 7 (802.11be) feature allowing simultaneous multi-band links.
- TR-181 — Broadband Forum data model standard used for device management (USP/TR-369, CWMP).
- HAL — Hardware Abstraction Layer.
- DFS — Dynamic Frequency Selection; radar-avoidance requirement for certain 5 GHz Wi-Fi channels.
- AFC — Automated Frequency Coordination; a database-driven mechanism required for Standard Power 6 GHz operation in some regions.
- LPI / VLP / SP — Low Power Indoor / Very Low Power / Standard Power; regional 6 GHz operating modes with different power limits and indoor/outdoor rules.
- FCC / ISED — United States Federal Communications Commission / Canada's Innovation, Science and Economic Development radio equipment certification bodies.
- RED / CE — EU Radio Equipment Directive / Conformité Européenne marking.
- MTCTE — India's Mandatory Testing and Certification of Telecommunication Equipment framework.
- ANATEL — Brazil's national telecommunications agency and certification scheme.
- PICS — Protocol Implementation Conformance Statement; a formal, itemized declaration of which specification requirements a specific implementation supports (Section 9).
- RFC 2119 — the IETF document defining the normative keywords MUST/SHOULD/MAY and their variants, used throughout this specification per Section 1.3.

9. PICS Framework

A Protocol Implementation Conformance Statement (PICS) is a formal, itemized declaration by an implementer of which requirements in this specification a specific hardware SKU supports. This section defines the framework for producing one — category ranges, identifier prefixes, and a machine-readable schema pointer — not a completed, pre-filled matrix; completing an actual PICS for a specific SKU means instantiating every individual identifier from Sections 3.4–4.5 (101 total — see Section 9.1) with that SKU's Supported/Not Supported/N/A status. Its purpose is to let a silicon vendor, OEM, or operator state conformance precisely enough for another party to assess interoperability

without re-deriving it from prose — the same function a PICS serves in IEEE, Broadband Forum, and Wi-Fi Alliance specifications.

9.1 PICS Category Summary

Every requirement in Sections 3–4 carries a stable identifier (Section 1.3). The table below summarizes the identifier ranges by category; a complete PICS declaration states support (Yes / No / Not Applicable) for each individual identifier within these ranges, not just the category as a whole.

Category	Prefix	ID Range	Count	Section	Companion JSON File
CPU/RAM/Flash Floor	CRF	001–002	2	Section 3	<profileId>.json
Compute Architecture	CA	001–012	12	Section 3.4	<profileId>.json
Memory	MEM	001–010	10	Section 3.5	<profileId>.json
Peripherals (top-level)	PER	001–009	9	Section 4	<profileId>.json
Platform-Control Hardware	PCH	001–020	20	Section 4.2	<profileId>.json
Flash Logical Classes	FLC	001–005	5	Section 4.3	flash-layout/<profileId>.flash-layout.json
Storage (extended)	STG	001–010	10	Section 4.3	flash-layout/<profileId>.flash-layout.json
Network Hardware	NET	001–015	15	Section 4.4	<profileId>.json
Wi-Fi Hardware	WIFI	001–018	18	Section 4.5	<profileId>.json

9.2 Completing a PICS

- For each requirement identifier in the ranges above, state Supported = Yes, No, or Not Applicable, plus any implementation-specific notes (e.g., a measured benchmark score for CA-003, or the exact watchdog timeout used for PCH-006).
- A "No" against any requirement whose tabular status in Sections 3–4 is "Required" is a nonconformance and should be flagged as such, not omitted.
- A PICS is filed per hardware SKU, not per profile in the abstract — two SKUs both claiming the EthWAN WiFi Router profile may have different PICS if their hardware differs (e.g., one includes Wi-Fi 7/MLO, the other does not).
- Regional conformance (Section 7) is declared separately, per region, using the regional.schema.json compliance array — it is not part of the core PICS in Section 9.1, since regional certification status changes independently (Section 7.5).

9.3 Machine-Readable PICS

The JSON companion files described in Section 3.2 provide inputs for generating a machine-readable PICS — each field corresponds to one or more requirement identifiers, but a completed PICS must additionally declare status (Supported/Not Supported/N/A) and evidence for every applicable requirement identifier (101 total, Section 9.1), which these files do not yet do individually per-ID. Representative examples:

JSON Field	Requirement ID	Note
cpu.minClockGHz, cpu.minCores	CRF-001/002	One CRF ID per profile in the primary hw-compat file.

JSON Field	Requirement ID	Note
peripherals.connectivity.wan.required	PER-001	Boolean maps to Supported = Yes/No directly.
peripherals.connectivity.wifi.requiredSecurity	WIFI-016	Array should include both "WPA2" and "WPA3" to declare WIFI-016 supported.
flash-layout partitions[] entries	FLC-001...005 / STG-001...010	Each partitions[] entry's sizeMB, once populated, evidences the corresponding FLC/STG identifier.
regional/<profileId>.<regionId>.json compliance[]	N/A — Section 7, not Section 9	Regional conformance is declared separately per Section 9.2 — not part of the core PICS identifier ranges above.

A fully machine-readable PICS requires a per-requirement declaration containing the requirement identifier, applicability, support status, evidence reference, and implementation notes. The current hw-compat and companion JSON files provide inputs for generating that declaration but do not constitute a completed PICS by themselves

9.4 PICS Declaration Header

Every completed PICS, whether tabular or JSON, should carry the following identifying information:

Implementer / Vendor	To Be Planned
Hardware SKU / Model	Banana Pi R4
Profile Claimed	EthWAN WiFi Router EXT EasyMesh
Specification Revision	R1
Date Completed	18 Sep 2026
Contact	Chandrakanth Pokuru